$$\frac{\partial u_c(\omega)}{\partial \omega} = gm\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{1}{m\omega-1}}\left(-\frac{m}{(m\omega-1)^2}\ln\frac{c}{(\pi m+r)\omega}-\frac{1}{(m\omega-1)\omega}\right)-cm - (\pi m+r)\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{m\omega}{m\omega-1}}\left(-\frac{m}{(m\omega-1)^2}\ln\frac{c}{(\pi m+r)\omega}-\frac{m}{m\omega-1}\right). \tag{35}$$

$$G(\omega) = \frac{c\left(1-m\omega-\ln\frac{c}{(\pi m+r)\omega}+(1-m\omega)^2\omega\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{1}{1-m\omega}}\right)}{1-m\omega-m\omega\ln\frac{c}{(\pi m+r)\omega}}. \tag{36}$$

sampling ratio ω decreases the worker's expected payoff due to the verification. Intuitively, this pushes the worker to adopt more honest behavior. There is a threshold value of ω where the worker's best response becomes fully trustworthy, i.e., $\beta^\star = 0$, once ω exceeds that threshold. On the other hand, if $r \le -(\pi - c)m$, fully trustworthy computing can never be the worker's best response for any ω. These regions are illustrated in Fig. 4.

From Theorem 1, if

$$\omega \ge \frac{1}{m} \quad \text{and} \quad r > -(\pi - c)m, \tag{29}$$

or if

$$0 < \omega < \frac{1}{m} \quad \text{and} \quad r \ge -\pi m + \frac{c}{\omega}, \tag{30}$$

then u_w is maximized at $\beta^\star = 0$, so the worker prefers trustworthy computation. Fig. 4 visualizes how the minimum required r grows as ω decreases under different β. The green area marked by $\beta^\star = 0$ corresponds to trustworthy computation, while the nonzero-$\beta^\star$ area indicates parameter choices where the worker has an incentive to cheat. The deterrence effect of sampling is not uniform across ω. In the low-sampling regime, i.e., $0 < \omega < 1/m$, the trust boundary is the curve $r = -\pi m + c/\omega$. Its steep shape near small ω indicates a strong substitution between sampling and escrow. Here, a small increase in ω can sharply reduce the deposit required to deter misbehavior. Fig. 4 also reveals a transition at the point $\omega = 1/m$. Once $\omega \ge 1/m$, the deposit threshold no longer decreases. In this case, a larger ω can still raise the detection probability for trustworthy computing, but it also increases the client's verification overhead.

B. Optimal Sampling Strategy

A client can select ω to obtain trustworthy service, i.e., to guarantee that the worker computes with $\beta = 0$. Given $r > -(\pi - c)m$, we establish a sufficient condition under which the client's payoff $u_c(\omega)$ is strictly increasing.

Theorem 2. *Suppose $r > -(\pi - c)m$ and $g > \pi$. If the payment-to-cost ratio satisfies*

$$\frac{\pi}{c} \ge \frac{3}{2}, \tag{31}$$

then, by using the optimal strategy, $u_c(\omega)$ is increasing in ω when the worker for $0 < \omega < c/(\pi m + r)$.

Proof. By Theorem 1, as the worker uses its best response β_0, the client's payoff becomes

$$u_c(\omega) = gm\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{1}{m\omega-1}} + r - \omega mc - (\pi m+r)\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{m\omega}{m\omega-1}}. \tag{32}$$

By differentiating the client's payoff with respect to ω, we can show that $\mathrm{d}u_c(\omega)/\mathrm{d}\omega > 0$ is equivalent to $g > G(\omega)$, where $\mathrm{d}u_c(\omega)/\mathrm{d}\omega$ and $G(\omega)$ are given in (35) and (36) respectively. Define the payment-to-cost ratio $\rho = \pi/c$, and the relevant $G(\omega) \le c\rho$ is a sufficient condition for $\mathrm{d}u_c(\omega)/\mathrm{d}\omega > 0$ since $g > \pi = c\rho$.

To establish this comparison, we derive an upper bound on $G(\omega)$. In the following proof, we first show that its denominator is positive and then upper-bound the nonlinear exponential term in its numerator. Let

$$x = \frac{(\pi m+r)\omega}{c}. \tag{37}$$

For $0 < \omega < c/(\pi m + r)$, we have $0 < x < 1$ and $m\omega = \frac{cm}{\pi m+r}x$. Using the inequality $x\ln(1/x) \le 1 - x$, we find that the denominator of $G(\omega)$ satisfies

$$1-m\omega-m\omega\ln\frac{c}{(\pi m+r)\omega} = 1-\frac{cm}{\pi m+r}x\left(1+\ln\frac{1}{x}\right) \ge 1-\frac{cm}{\pi m+r} > 0, \tag{38}$$

where the last inequality follows from $r > -(\pi - c)m$.

Next, we consider the term in the numerator of $G(\omega)$:

$$(1-m\omega)^2\omega\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{1}{1-m\omega}} = \frac{c(1-m\omega)^2}{\pi m+r}\exp\left(\frac{m\omega}{1-m\omega}\ln\frac{c}{(\pi m+r)\omega}\right), \tag{39}$$

where the exponent is nonnegative and can be bounded as

$$0 \le \frac{m\omega}{1-m\omega}\ln\frac{c}{(\pi m+r)\omega} = \frac{\frac{cm}{\pi m+r}x\ln(1/x)}{1-\frac{cm}{\pi m+r}x} \le \frac{\frac{cm}{\pi m+r}(1-x)}{1-\frac{cm}{\pi m+r}x} \le \frac{cm}{\pi m+r} \le \frac{1}{\rho}. \tag{40}$$

$$\left(1-\rho\frac{cm}{\pi m+r}x-\frac{2c^2m}{(\pi m+r)^2}x(1-\frac{cm}{\pi m+r}x)\right)\ln\frac{1}{x}\geq\left(\frac{cm}{\pi m+r}x-1\right)\left(\rho-1-\frac{c}{\pi m+r}(1-\frac{cm}{\pi m+r}x)\right). \quad (46)$$

$$\left(1-\frac{cm}{\pi m+r}x\right)\left(\rho-1-\frac{cm}{\pi m+r}(1-\frac{cm}{\pi m+r}x)\right)+\left(1-\rho\frac{cm}{\pi m+r}x-2(\frac{cm}{\pi m+r})^2x(1-\frac{cm}{\pi m+r}x)\right)\ln\frac{1}{x}\geq 0. \quad (47)$$

The first inequality follows from $x\ln(1/x)\leq 1-x$. The second follows from $cm/(\pi m+r)<1$, while the last one follows from $r\geq 0$. We now use the relaxation

$$e^z=1+z+\sum_{k=2}^{\infty}\frac{z^k}{k!}\leq 1+z+\frac{1}{2}\sum_{k=2}^{\infty}z^k \quad (41)$$

$$=1+z+\frac{z^2}{2(1-z)}\leq 1+2z,\quad \forall z\in\left(0,\frac{2}{3}\right], \quad (42)$$

where e^z is no larger than $1+2z$ whenever $z^2/(2(1-z))\leq z$, which is equivalent to $z\leq 2/3$. Since the exponent in (39) is at most $1/\rho$, this relaxation holds uniformly over the entire low-sampling region whenever

$$\rho\geq\frac{3}{2}. \quad (43)$$

Then, we have

$$(1-m\omega)^2\omega\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{1}{1-m\omega}} \leq\frac{c(1-m\omega)^2}{\pi m+r}\left(1+\frac{2m\omega}{1-m\omega}\ln\frac{c}{(\pi m+r)\omega}\right). \quad (44)$$

Finally, we upper-bound $G(\omega)$ using (44). Since its denominator is positive, substituting this numerator bound shows that $G(\omega)/c\leq\rho$ holds provided that

$$1-m\omega-\ln\frac{c}{(\pi m+r)\omega} +\frac{c(1-m\omega)^2}{\pi m+r}\left(1+\frac{2m\omega}{1-m\omega}\ln\frac{c}{(\pi m+r)\omega}\right) \leq\rho\left(1-m\omega-m\omega\ln\frac{c}{(\pi m+r)\omega}\right). \quad (45)$$

Substituting $x=(\pi m+r)\omega/c$ and rearranging, (45) is equivalent to (46). Since $m\geq 1$, a sufficient condition is (47). We now verify (47) by obtaining a positive lower bound for its left-hand side (LHS). The coefficient of ρ is

$$1-\frac{cm}{\pi m+r}x\left(1+\ln\frac{1}{x}\right)>0, \quad (48)$$

and therefore replacing ρ by $3/2$ gives a lower bound. At $\rho=3/2$, both the constant term and the coefficient of $\ln(1/x)$ decrease with $cm/(\pi m+r)$ on $(0,2/3]$. Since $\ln(1/x)>0$ and $cm/(\pi m+r)\leq 2/3$, the LHS of (47) is therefore bounded below by

$$\frac{(3-2x)(8x-3)}{54}+(1-x)\ln\frac{1}{x}-\frac{8}{9}\left(1-\frac{2x}{3}\right)x\ln\frac{1}{x}. \quad (49)$$

By keeping the positive and negative logarithmic terms in (49) separate, we use the following bounds:

$$\ln\frac{1}{x}\geq 1-x,\qquad x\ln\frac{1}{x}\leq\frac{1-x^2}{2},\qquad 0<x<1, \quad (50)$$

where the second follows by integrating $1/t\leq\left(1+t^{-2}\right)/2$ from x to 1. Applying these bounds to the two logarithmic terms gives

$$\text{LHS of (47)} \geq\frac{(3-2x)(8x-3)}{54}+(1-x)^2-\frac{4}{9}\left(1-\frac{2x}{3}\right)(1-x^2) =\frac{14\left(x-\frac{1}{2}\right)^2+16(1-x)^3+\frac{3}{2}}{54}>0. \quad (51)$$

Consequently, whenever $\pi/c\geq 3/2$, we have $G(\omega)\leq c\rho=\pi<g$ for every $0<\omega<c/(\pi m+r)$, and

$$\frac{\partial u_c(\omega)}{\partial\omega}>0,\qquad 0<\omega<\frac{c}{\pi m+r}. \quad (52)$$

Thus, $\pi/c\geq 3/2$ is sufficient for the client's payoff to be strictly increasing throughout the low-sampling region, which completes the proof. □

Theorem 2 establishes that, under the condition of $\pi/c\geq 3/2$, the client's utility increases throughout the low-sampling region $0<\omega<c/(r+\pi m)$. At $\omega=c/(r+\pi m)$, the worker's best response switches to trustworthy computation. Beyond this boundary, correctness is already ensured such that additional sampling only increases the client's verification cost. This transition determines the client's optimal sampling strategy, as we formalize in the following theorem.

Theorem 3. *Given $r>-(\pi-c)m$ and $\pi/c\geq 3/2$ satisfying Theorem 2, the client's optimal sampling ratio is*

$$\omega^\star=\frac{c}{r+\pi m}. \quad (53)$$

Proof. By Theorem 2, u_c is strictly increasing with ω on $0<\omega<c/(r+\pi m)$. For $\omega\geq c/(r+\pi m)$,

$$u_c=gm-\pi m-\omega mc, \quad (54)$$

and we have $\frac{\partial u_c}{\partial\omega}=-mc<0$. Hence the maximum occurs at $\omega=c/(r+\pi m)$ with

$$u_c\big|_{\omega=\frac{c}{r+\pi m}}=gm-\pi m-\frac{mc^2}{r+\pi m}, \quad (55)$$

which proves that (53) is the unique optimum and completes the proof. □

Theorem 3 provides an optimal strategy for choosing the sampling ratio ω under the conditions $r>-(\pi-c)m$ and $\pi/c\geq 3/2$. The latter sufficient condition is required only for the monotonicity argument in Theorem 2. Once the deposit r and price π are fixed, for $\omega<\omega^\star$, extra samples increase both the chance of catching errors and the expected deposit refund, whereas for $\omega>\omega^\star$, more samples add verification cost and reduce the client's profit. From (53), we can see that larger r